

THE POWER TO PRESS : EVALUATING PAIN, DOSE, AND IMAGE QUALITY IN PATIENT-LED MAMMOGRAPHY

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BACKGROUND

Mammography is the gold standard for early breast cancer detection, but compression-associated pain severely hinders screening adherence. While self-assisted compression may enhance the patient experience, its potential impact on mammographic image quality remains unclear, highlighting a gap in evidence regarding whether patient-centered compression can be achieved without compromising diagnostic image quality.

OBJECTIVES

- Evaluate self-compression and its correlation with pain, radiation dose and image quality.
- Compare the pain intensity, radiation dose and image quality between self- and standard compression techniques.
- Analyse the association between self-compression with pain intensity, radiation dose and image quality.

METHODS

Quasi-experimental study in a tertiary centre in Kuantan, Pahang involving women undergoing screening and diagnostic mammography between December 2023 until November 2025.

412 participants (standard, n = 210; supervised self-assisted, n = 202). Excluded pregnant, breastfeeding, history of breast augmentation, inability to stand independently and mental inability.

Pain intensity categorized according to Defence and Veterans Pain Rating Scale (DVPRS) as none, mild, moderate and severe. Radiation dose evaluated using Mean Glandular Dose (MGD). Image quality graded via standardized Perfect-Good-Moderate-Inadequate (PGMI) criteria.

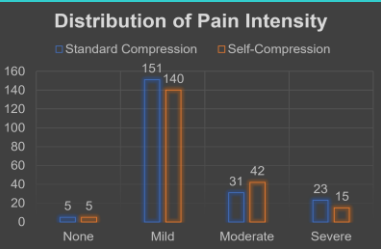
Data analyzed using Chi-square tests, Mann-Whitney U tests, multinomial logistic regression, and multiple linear regression to determine independent predictors of outcomes.

RESULTS

TABLE I: Who were our participants?

	Total sample (N = 412)	Standard Compression (n = 210)	Self- Compression (n = 202)
Age Group			
31 – 50 years	186 (45%)	102 (49%)	84 (42%)
51-60 years	129 (31%)	64 (30%)	65 (32%)
>60 years	97 (24%)	44 (21%)	53 (26%)
Ethnicity			
Malay	299 (73%)	155 (74%)	144 (71%)
Chinese	86 (21%)	41 (20%)	45 (22%)
Indian/ Others	27 (6%)	14 (7%)	13 (6%)
Education Level			
Secondary or Below	203 (49%)	99 (47%)	104 (51%)
Tertiary or Higher	209 (51%)	111 (53%)	98 (49%)

FIGURE I: 71% experienced mild pain. Different compression, same pain (p = 0.308)



What predicts pain during mammogram?
Ethnicity & education level

What factors affect image quality during mammogram?
Younger age & ethnicity

What determines MGD and breast thickness?
Age, ethnicity & education level

KEY INSIGHT: Patient Factors
Matter More than Technique

Key numerical findings

- Mild pain for most (71%). Different compression, similar pain (p = 0.308).
- Dose-safe, regardless of technique (p > 0.05).
- Image quality maintained with self-compression, despite significant association with compression techniques (p = 0.024).
- Pain intensity is significantly predicted by ethnicity ($\chi^2 = 15.793$, p = 0.015) and education level (p = 0.001).
- Good image quality is significantly associated with younger age (OR = 2.910, CI: 1.278-6.626, p = 0.011) and ethnicity (p < 0.001).
- MGD is significantly predicted by age (p < 0.001), ethnicity (p = 0.028) and education level (p < 0.001).
- Breast thickness is significantly predicted by age (p = 0.008), ethnicity (p = 0.005) and education level (p < 0.001).

	Standard Compression	Self- Compression	P-value
Breast Thickness*			
Median on CC (mm)	49	50	0.103
Median on MLO	56	57	0.081
Mean Glandular Dose*			
Median on CC	0.9	0.9	0.897
Median on MLO	1.1	1.0	0.480
Image Quality**			
Good	93	67	$\chi^2 = 7.488$ p = 0.024
Moderate	106	114	
Inadequate	11	21	

* Mann-Whitney U test
** Pearson chi-square test

TABLE II: Clinical Reassurance
Dose-Safe, Regardless of Technique
Quality Maintained with Self-Compression

CONCLUSION / TAKE-HOME MESSAGE

Patients can take control without compromising diagnostic adequacy.

Self-assisted compression offers comparable pain and radiation dose, with predominantly moderate-to-good image quality.

The patient, not the compression technique; shape the screening experience.

Sociodemographic factors have greater influence than who controlled compression.

Self-assisted compression as a patient-centred option in mammography practice.

Further research should evaluate implementation across diverse populations and screening settings.

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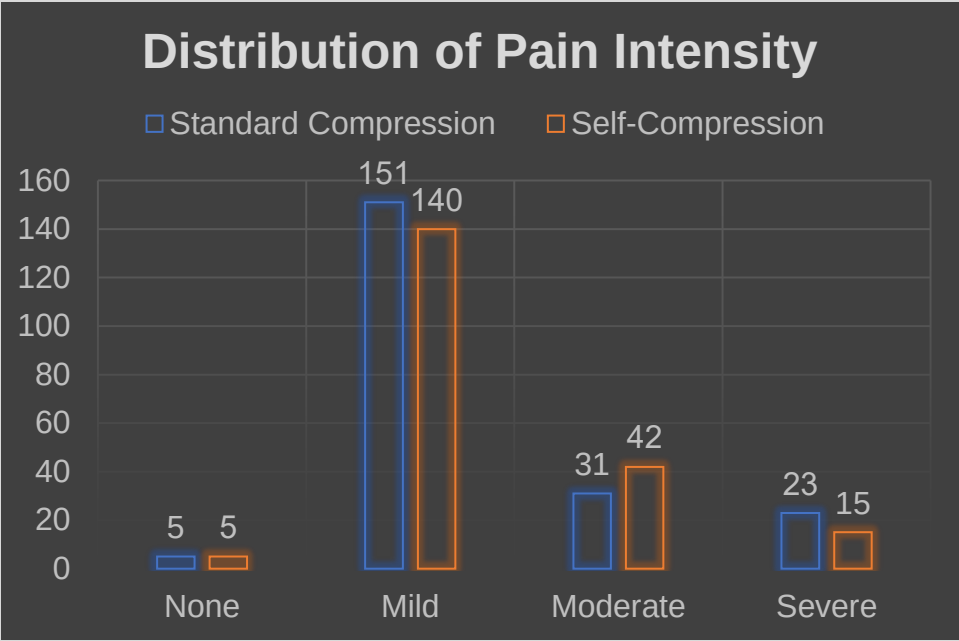
ACKNOWLEDGEMENTS

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No significant difference in pain intensity between compression techniques ($p = 0.308$).

Image quality	Standard	Self-assisted
Poor		
Moderate-Good		
Total	N (100%)	N (100%)
χ^2 /Fisher's exact test: p = 0.024		

Self-assisted compression achieved predominantly moderate-to-good image quality, indicating acceptable diagnostic adequacy despite a significant difference in image-quality distribution (p = 0.024).

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